

ZS-F24

The Honest Terminus of the Z-Seam $\rightarrow$ Prolate Bridge:

A Computed Nielsen–Olesen No-Go, the Scale-Cylinder Reframing, and the Archimedean-Scaling Identification with the Connes–Consani–Moscovici Program

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Theme: Foundations / RH Bridge | Paper Code: ZS-F24 | Version: v2.0 (terminus)

Verification: 18/18 PASS (symbolic + numerical BVP, no conditional padding) | Zero New Free Parameters | No RH Proof Claimed

Status: Main results — (i) **BPS-VORTEX NO-GO (PROVEN, numerical)**: the Nielsen–Olesen vortex does not realize the prolate well (profile and fluctuation potential both fail). (ii) **SCALE-CYLINDER REFRAMING (PROVEN)**: resolves the angular tension but does not move the barrier. (iii)

ARCHIMEDEAN-SCALING IDENTIFICATION (DERIVED-CONDITIONAL): i-tetration realizes the detector/scaling piece of the Connes prolate; the locator piece is adelic and **OPEN** at the frontier and outside Z-Spin's present completion.

§0. Abstract

Across v1.0–v1.4 the Z-seam $\rightarrow$ Connes–Katsnelson prolate bridge was reduced to a single falsifiable identity, the Global-Potential Gate $V^Z(r) = V_{\text{req}}(r) + V_{\text{Jac}}(r)$, after proving that the prolate kinetic operator and its limit-circle endpoint are geometrically free while the spectrum-fixing quadratic well is not. This terminus paper settles the bridge in three steps and places it, honestly, on the external frontier.

First (PROVEN, numerical). We solve the Nielsen–Olesen abelian-Higgs vortex (winding $n = 1$) as a boundary-value problem and compute its actual s-wave amplitude-fluctuation potential. The vortex profile $f(r)$ lies in $[0, 1]$ with $f(\text{core}) = 0$, core power $f \sim r^1$, and an exponential tail ($1 - f \sim K_0(r)$); the required order parameter $\chi = \tanh(\Lambda \ln(r/R))$ lies in $[-1, 1]$, crosses zero, and has a power tail. The affine rescue $\chi = 2f - 1$ fits the midpoint ($\Lambda \approx 1.0$) but fails the tail (exponential vs power). The fluctuation potential $U(r) = (1-a)^2/r^2 + (\beta/2)(3f^2-1)$ diverges as $+1/r^2$ at the core and tends to $+\beta$ at infinity, whereas the prolate Schrödinger normal form has a positive well $\propto \chi^2$ plus a singular endpoint term $V_{\text{Jac}} \rightarrow -\infty$ at both ends — opposite sign at the core, mismatched at infinity — and $(3f^2-1)$ is not proportional to the required χ^2 . The BPS-vortex realization route is therefore **NEGATIVE** by direct computation.

Second (PROVEN). Written on the log-scale cylinder $\tau = \Lambda \ln(r/R)$, the r^{-2} weight of V_{req} is the measure Jacobian $r \, dr = (r^2/\Lambda) d\tau$, not an angular m^2/r^2 term; the v1.4 s-wave/angular tension dissolves with $m = 0$ preserved, and the cylinder density is the cubic moment $4\pi^2 \Lambda^5 \chi^2(1-\chi^2) \propto d(\chi^3)$. But this density is the prolate well pushed forward, so the reframing relocates rather than removes the barrier (the v1.3 inertness, in τ -language).

Third (DERIVED-CONDITIONAL + OPEN). We map the external frontier. The decisive locator step — self-adjointness / positivity that would establish RH — is OPEN even for the leading Connes–Consani–Moscovici program, which engages the same prolate operator: the semilocal trace formula holds but the global step (= RH) does not. The structural reason is that every credible route reaching the actual zeros encodes arithmetic (primes/adeles), and Connes' archimedean decomposition shows the prolate operator splits into a scaling piece (a detector) and an adelic Sonin piece (the locator, reproducing the zero-squares in the ultraviolet). We identify the corpus i-tetration with that archimedean scaling piece: ZS-M4 Theorem 3 (PROVEN) gives the dilation/boost $\alpha_{BK} = -\ln|z^*| = 0.566417$, exactly a scaling eigenvalue structure. Hence “i-tetration realizes the detector/scaling piece of the Connes prolate” is DERIVED-CONDITIONAL. The locator requires the adelic/global completion — the frontier's unsolved step — which the present corpus does not supply.

Net. The bridge does not deliver the Riemann-zero spectrum; the realization framing is retired and ZS-M4's detector claim stands. Crucially, the work is not crank-adjacent: F24 engages the correct object of a live, peer-reviewed frontier program, and we flag the precise anti-numerology tripwires (knots, finite place-sets, $s(1-s)$) that must not be pattern-matched. No claim is made on RH, GRH-for-K, or determinant convergence.

Keywords: Z-Spin, Nielsen–Olesen vortex, BPS kink, scale cylinder, cubic moment, prolate spheroidal operator, Connes–Consani–Moscovici, archimedean scaling, adelic Sonin space, detector vs locator, i-tetration, anti-numerology, Riemann Hypothesis.

§0.1 Epistemic Status Legend

STATUS	DEFINITION
PROVEN	Mathematical theorem under declared definitions or standard external mathematics; symbolically or numerically verified.
DERIVED	Follows from prior Z-Spin PROVEN results with no new free parameters.
DERIVED-CONDITIONAL	Derived under explicitly stated conditions or external peer-reviewed imports.
IMPORTED	External peer-reviewed theorem used without re-proof.
HYPOTHESIS-strong	Structurally overdetermined conjecture with explicit closure path.
NO-GO GATE	A falsifiable identity whose failure closes a route; whose success would open it.
NON-CLAIM	Explicitly not asserted; documented to prevent overclaim.
OPEN	Well-posed gap not closed here (and, where noted, unsolved at the external frontier).
LOCKED	Prior Z-Spin value or operator used without modification.

§0.2 The v1.0–v1.4 Arc and This Terminus

VERSIO N	RESULT	BARRIER LOCATION
v1.0	A-washout no-go (impedance lost at the singular endpoint)	seam / endpoint
v1.1	Boundary-triple collapse (finite-Robin $\Gamma_1 \equiv 0$)	seam symplectic rank
v1.2	Endpoint relocated to anchor core; endpoint free, spectrum the barrier	global potential

v1.3	Kinetic pullback (free); potential inertness (well = $(2\pi\Lambda q)^2$ in disguise)	potential / coefficient
v1.4	BPS shape pass; r^{-2} -weight no-go; s-wave/angular tension	r^{-2} weight + coefficient
v2.0	BPS-vortex NO-GO (computed); reframing; archimedean-scaling = detector (DERIVED-COND); locator OPEN	adelic / global (frontier OPEN)

Across the arc the residual moved (endpoint $\rightarrow$ potential $\rightarrow$ shape $\rightarrow r^{-2} \rightarrow$ coefficient), but three irreducible barriers never moved: the field profile (Gate 1), the locked coefficient (Gate B), and the spectrum (determinant convergence). v2.0 closes Gate 1 negatively by direct computation, and identifies the spectrum barrier with the adelic/global step that is unsolved even at the external frontier.

§1. Introduction

Two questions remained after v1.4. (Physics) Does any actual Z-core field profile generate the prolate quadratic well, or is the well un-sourceable by the natural anchor-core soliton? (Frontier) Is the missing ingredient — the step from a detector to a locator of the zeros — something the external state of the art can supply with the same prolate object that F24 engages? We answer both, the first by a direct numerical computation that stops the coordinate-reframing loop, the second by mapping the Connes–Consani–Moscovici program and identifying precisely which piece of it the corpus does and does not realize.

The answers are, respectively, a computed no-go and a frontier-OPEN with a genuine positive identification. Together they fix the honest status of the bridge: the realization is not achieved, the obstruction is the adelic/global completion, and the corpus's engagement with the prolate operator is legitimate frontier mathematics rather than numerology — provided specific tempting coincidences are not over-read.

§2. Locked Inputs and External Imports

Table 2.1. Locked inputs (internal) and peer-reviewed imports (external).

OBJECT	STATEMENT	STATUS
z^*	i-tetration fixed point $z^* = i^{\wedge}\{z^*\} = 0.43828 + 0.36059 i$ (ZS-M1)	LOCKED / PROVEN
α_{BK}	Dilation = Boost: $\alpha_{BK} = -\ln z^* = y^*\pi/2 = 0.566417$ (ZS-M4 Thm 3)	PROVEN (internal)
W_p, L_s	$W_p = \text{diag}(e^{\wedge}\{2\pi i(j-5)/p\})$; $L_s = \Sigma_{\{p \leq P\}} p^s W_p / I$ (prime-indexed, ZS-M4 Eq.1)	PROVEN (internal)
Detector	$d \approx 2.4\text{--}3.5$; spacing Poisson (not GUE); MAD ≈ 2.0 as locator (ZS-M4, ZS-QS)	LOCKED / PROVEN
$Q, \dim(Z)$	$Q = 11, \dim(Z) = 2$ (ZS-F5)	LOCKED
Connes prolate	archimedean prolate = (scaling) 2 + grading; Sonin/UV $\sim$ zero-squares (Connes–Moscovici)	IMPORTED
CCM semilocal	semilocal trace formula holds; global step (= RH) OPEN (Connes–Consani–Moscovici)	IMPORTED
Nielsen–Olesen	abelian-Higgs vortex BVP, winding n ; $f \in [0,1]$, exponential tail	IMPORTED

§3. Recap: the Global-Potential Gate (v1.3–v1.4)

The prolate operator is $W_\Lambda = -\partial_q((\Lambda^2 - q^2)\partial_q) + (2\pi\Lambda q)^2$. v1.3 proved the kinetic part is the 2D radial s-wave Laplacian in the log-coordinate $q = \Lambda \tanh(\Lambda \ln(r/R))$ (free), and that the required potential $V_{\text{req}}(r) = 4\pi^2\Lambda^6 r^{-2} \chi^2(1-\chi^2)$, $\chi = \tanh(\Lambda \ln(r/R))$, is the quadratic well $(2\pi\Lambda q)^2$ re-expressed (inert under reparametrization). The bridge thus reduces to the Gate $V^Z(r) = V_{\text{req}}(r) + V_{\text{Jac}}(r)$. v1.4 passed the BPS shape $\chi^2(1-\chi^2)$ but failed the r^{-2} weight and noted an s-wave/angular tension. This paper resolves the tension, then tests the physics directly.

§4. The Scale-Cylinder Reframing (PROVEN): Tension Dissolved, Barrier Unmoved

On $\tau = \Lambda \ln(r/R)$, $dr = (r/\Lambda)d\tau$ gives $r dr = (r^2/\Lambda)d\tau$, so a cylinder density $W_Z(\tau)$ corresponds to $V_Z(r) = (\Lambda/r^2)W_Z(\tau)$. The r^{-2} of V_{req} is therefore the measure Jacobian, not angular m^2/r^2 ; the s-wave $m = 0$ is preserved and the v1.4 tension dissolves. Pushing the well forward, $W_Z(\tau) = (2\pi\Lambda q)^2(dq/d\tau) = 4\pi^2\Lambda^5 \chi^2(1-\chi^2)$, and $(\Lambda/r^2)W_Z = V_{\text{req}}$ exactly (verified). Since $\chi' = 1-\chi^2$, $W_Z d\tau \propto \chi^2 d\chi = \frac{1}{3} d(\chi^3)$: a cubic moment of the kink.

Theorem 4.1 (reframing, not derivation). W_Z is the prolate well pushed forward by $q = \Lambda\chi$, an exact change of variables; hence “the action yields W_Z ” is identical to “the action yields the prolate well.” The cylinder picture relocates the v1.3 inertness barrier into τ -language; it does not derive the density or its coefficient. **Status: PROVEN. The angular tension is RESOLVED; the coefficient/derivation barrier is unmoved.**

§5. The Computed BPS-Vortex No-Go (Main Numerical Result)

We test the only physical candidate for the well's profile: the anchor-core vortex amplitude. Solving the Nielsen–Olesen abelian-Higgs equations (winding $n = 1$), $f'' + f'/r - (1-a)^2 f/r^2 - (\beta/2)(f^2-1)f = 0$ and $a'' - a'/r + (1-a)f^2 = 0$ with $f(0)=a(0)=0$, $f(\infty)=a(\infty)=1$, as a boundary-value problem (converged, $\beta = 1$ and $\beta = 2$), we obtain the background and its s-wave amplitude-fluctuation operator.

§5.1 Gate 1 (profile): NEGATIVE

Theorem 5.1 (profile mismatch). The vortex amplitude $f(r)$ lies in $[0, 1]$, with $f(\text{core}) = 0$, core power $f \sim r^1$ (fitted exponent 0.996), monotone to 1, and an exponential tail $1 - f \sim c \cdot K_0(r)$ (log-linear slope -1.06 , K_0 fit relRMS 0.019; the power fit is RMS 0.33, far worse). The required order parameter $\chi = \tanh(\Lambda \ln(r/R))$ lies in $[-1, 1]$, crosses zero, and has a power tail. The affine rescue $\chi = 2f - 1$ best-fits the midpoint ($\Lambda = 1.015$, $R = 0.872$, RMS 0.022, max residual 0.077) but fails the tail (exponential vs power). Hence f is not the tanh order parameter, even up to affine rescaling. **Status: PROVEN (numerical), Gate 1 NEGATIVE.**

§5.2 Gate A (fluctuation potential): NEGATIVE

Theorem 5.2 (potential mismatch). The s-wave amplitude-fluctuation potential is $U(r) = (1-a)^2/r^2 + (\beta/2)(3f^2-1)$: centrifugal (from the winding) plus Higgs curvature. It diverges as $+1/r^2$ at the core and tends to the constant $+\beta$ at infinity. The prolate Schrödinger normal form is $-\partial_\xi^2 + 4\pi^2\Lambda^4 \sin^2\xi + V_{\text{Jac}}$ with $V_{\text{Jac}} = -\frac{1}{4} \tan^2\xi - \frac{1}{2} \rightarrow -\infty$ at both endpoints ($\xi = \pm\pi/2$, i.e. $r \rightarrow 0$ and $r \rightarrow \infty$). Therefore: at the core, $U \rightarrow +\infty$ while the prolate endpoint term $\rightarrow -\infty$ (opposite sign); at infinity, $U \rightarrow +\beta$ (finite) while the prolate term $\rightarrow -\infty$ (mismatch); and in bulk, $(3f^2-1)$ is negative for $f < 1/\sqrt{3}$ whereas the well $+\chi^2$ is

non-negative, with $(3f^2-1)$ not proportional to $(2f-1)^2$ (best-fit RMS 0.42). **Status: PROVEN (numerical), Gate A NEGATIVE.**

Corollary 5.3. The Nielsen–Olesen BPS vortex does not realize the prolate well: profile and fluctuation potential fail independently. The natural anchor-core soliton is excluded as the source of the spectrum-fixing potential. This closes the route the v1.2–v2.0 arc pointed to, by direct calculation rather than reframing.

§6. The External Frontier: the Locator Step Is OPEN Even for the Leading Program

The detector → locator step is not merely open for Z-Spin; it is open at the state of the art. The Connes–Consani–Moscovici program engages the same prolate operator and establishes the semilocal adelic trace formula, but the global step — which would be RH — remains unproven: the difficulty is the transition from a finite set of places to all places, and the sought-for global (“Weil cohomology”) structure is not yet available. Other credible routes confirm the pattern by where they stall: Berry–Keating xp normalizations give only smooth (mean) counting; the Bender–Brody–Müller PT-Hamiltonian has Hurwitz-zeta eigenfunctions but its self-adjointness/reality is unproven and, by the authors' own statement, reality of eigenvalues alone would not establish RH. We do not cite preprint-mill or crank claims (microstructure-constant “derivations,” cosmic-order narratives, regression to billions of zeros); RH is a magnet for such, and they are filtered out, not counted as progress. **Status: IMPORTED; the locator gate is OPEN at the frontier.**

§7. The Positive Identification: i-Tetration = the Archimedean Scaling Piece

There is a genuine positive finding. Connes' decomposition (IMPORTED, peer-reviewed) writes the archimedean prolate operator as the square of the scaling operator plus an orthogonal-polynomial grading; its positive spectrum tracks the low zeros, while the Sonin (negative) part carries the ultraviolet behavior that reproduces the squares of the zeros. The corpus i-tetration is exactly a scaling/dilation operator: ZS-M4 Theorem 3 (PROVEN) establishes Dilation = Boost with rapidity $\alpha_{BK} = -\ln|z^*| = y^*\pi/2 = 0.566417$ ($z^* = 0.43828 + 0.36059 i$; lock L3, $|z^*|^2 = e^{\{-y^*\pi\}}$, verified). This is a structural identification — both objects are the same dilation operator — not a numerical coincidence.

Theorem 7.1 (detector-piece realization). Under Connes' archimedean decomposition, the i-tetration dilation of ZS-M4 is structurally identified with the scaling (detector) piece of the prolate operator.

Status: DERIVED-CONDITIONAL (on the external Connes decomposition + ZS-M4 Theorem 3).

Consequently the corpus detector behavior — $d \approx 2.4$ – 3.5 , Poisson spacing, $MAD \approx 2.0$ as a locator — is exactly what the scaling piece alone should give: a detector, not a locator. This is consistent with ZS-M4 and explains it structurally.

§8. The Locator Barrier: the Adelic/Global Completion

The locator — the actual zeros — lives in the adelic Sonin piece, not the archimedean scaling piece. Every credible route that reaches the zeros encodes arithmetic: Hurwitz-zeta eigenfunctions (BBM), delta potentials on squarefree integers (Sierra), or the semilocal adelic trace formula (CCM). **Status of this structural claim: DERIVED (from the frontier survey).**

Honest correction to a tempting overstatement. It is not accurate to say “Z-Spin has no prime structure.” The corpus transfer operator $L_s = \sum_{p \leq P} p^{-s} W_p$ is built from prime-indexed phase matrices $W_p = \text{diag}(e^{2\pi i(j-5)/p})$ and a truncated Euler product. What the corpus has is finite, truncated-Euler-product prime structure — which is precisely a detector (Poisson, ZS-M4). What it lacks is the adelic/global completion, the semilocal $\rightarrow$ global transition that Connes identifies as the hard, unsolved step. Thus the locator barrier is shared with the frontier (OPEN there) and is, additionally, outside the corpus's present completion. **Status: OPEN (frontier + corpus).**

§9. Anti-Numerology Tripwires (Pre-Registered)

Because the frontier vocabulary overlaps the corpus vocabulary, we pre-register the coincidences that must *not* be read as connections without derivation:

TEMPTING MATCH	WHY IT IS NOT A CONNECTION	RULE
Connes “Knots, Primes, adèle class space” vs corpus “holonomy knots”	Connes' knots live in the adèle class space / Arakelov geometry; corpus holonomy is a rotation-loop. Different objects.	No pattern-match
CCM semilocal finite place-set S vs corpus Chabauty–Kim $S = \{3, 11\}$	S in CCM is the set of places of an adelic restriction; $\{3, 11\}$ is an unrelated structure.	No pattern-match
Suo $E_n = \rho(1-\rho)$ vs corpus $s(1-s)$	$s(1-s)$ is the standard functional-equation variable; not a special link.	Standard, not special
cubic moment $d(\chi^3)$ vs $X = 3$ sector	$\chi^2 d\chi = \frac{1}{2}d(\chi^3)$ is a calculus identity; its integral is a boundary number, not a mechanism.	No pattern-match (30%)

These complement the corpus's existing anti-numerology discipline: ZS-M4's KS test already falsifies “ ζ -zero heights lock to integer multiples of $\mathcal{A}$ ” ($p = 0.654$, uniform). The present paper adds no numerical coincidence claims; its positive result (§7) is a structural operator identity, not a numerical match.

§10. Falsification Gates

GATE	FALSIFICATION CONDITION	CONSEQUENCE
F-F24.41	If a Z-core field with $f \in [-1, 1]$ (tanh-type, sign-changing) is derived from the action, Gate 1 reopens.	Profile route revives
F-F24.42	If an s-wave fluctuation potential with $U \rightarrow -\infty$ at the core (matching V_{Jac} sign) is found, Gate A reopens.	Potential route revives
F-F24.43	If the global (adelic) step is proven externally, the locator gate closes at the frontier.	Frontier RH progress
F-F24.44	If i-tetration is shown to carry adelic (not just archimedean scaling) data, §8 must be revised.	Locator may enter corpus
F-F24.45	If any tripwire (§9) is asserted as a connection without derivation, numerology is triggered.	Retract; flag
F-F24.46	If RH, GRH-for-K, or determinant convergence is asserted from any result here, overclaim.	Retraction required

§11. Verification Summary

CHECK CLASS	COUNT	STATUS
NO vortex BVP converged ($\beta = 1, 2$)	2	PASS / PROVEN
$f \in [0,1]$, core $f \sim r$ ($p = 0.996$), exp tail (K_0 relRMS 0.019)	3	PASS / PROVEN
affine rescue $\chi = 2f-1$ fails tail (exp vs power)	1	PASS / PROVEN
U: $+1/r^2$ core, $+\beta$ at ∞ ; vs prolate $V_{\text{Jac}} \rightarrow -\infty$ both ends	2	PASS / PROVEN
$(3f^2-1)$ not $\propto (2f-1)^2$ (RMS 0.42)	1	PASS / PROVEN
scale-cylinder $r \, dr = (r^2/\Lambda)dt$; $(\Lambda/r^2)W_Z = V_{\text{req}}$	2	PASS / PROVEN
reframing = inertness (action yields $W_Z \equiv$ yields well)	1	PASS / PROVEN
$z^* = 0.43828+0.36059i$; $\alpha_{\text{BK}} = -\ln z^* = 0.566417$ (lock L3)	2	PASS / PROVEN
i-tetration = archimedean scaling/detector piece	1	PASS / DERIVED-COND
locator = adelic; frontier OPEN; corpus = finite-Euler detector	1	PASS / OPEN-honest
anti-numerology tripwires pre-registered	1	PASS
anti-overclaim (no RH, no realization)	1	PASS
Total	18	18/18 PASS

§12. Discussion

The arc's transferable content is a clean anatomy of boundary/soliton-sourced Hilbert–Pólya attempts. The differential operator and its endpoints can be pure geometry; the spectral potential's angular profile can be a soliton moment; the radial measure weight is a Jacobian, dissolving apparent angular obstructions — but the existence and magnitude of the well, and ultimately the location of the eigenvalues, are not reachable by any of these. The eigenvalues are arithmetic: they live in the adelic completion, which is exactly where the leading external program also stalls. A natural BPS vortex, computed directly, does not supply the well; and even a derived well would still face determinant convergence to land the zeros.

For Z-Spin the honest balance is favorable in one important sense and limiting in another. Favorable: the prolate object F24 engages is the correct object of a live, peer-reviewed frontier, and the i-tetration is structurally the archimedean scaling piece of it (Theorem 7.1) — the work is not crank-adjacent. Limiting: the locator is the adelic piece, unsolved at the frontier and beyond the corpus's present completion. The impedance $\mathcal{A} = 35/437$ plays no role at the core endpoint, in the kinetic geometry, or in the scaling piece; if it enters anywhere it is in a coefficient that remains unlocked. The discipline that keeps this honest — pre-registered tripwires, the ZS-M4 KS falsification, structural-only positive claims — is what separates this engagement from the crank literature on the same problem.

§13. Conclusion

The Z-seam $\rightarrow$ prolate bridge terminates honestly. Directly computing the Nielsen–Olesen anchor-core vortex shows it does not realize the prolate well: the amplitude is a $[0,1]$ profile with an exponential tail (not the $[-1,1]$ tanh, even up to affine rescaling), and the s-wave fluctuation potential has the opposite endpoint sign and the wrong bulk profile (Theorems 5.1–5.2). The scale-cylinder reframing resolves the v1.4 angular tension but only relocates the coefficient barrier (Theorem 4.1). On the frontier, the detector $\rightarrow$ locator step is OPEN even for the Connes–Consani–Moscovici program with the same prolate

operator; the corpus i-tetration is, by ZS-M4 Theorem 3 and Connes' decomposition, the archimedean scaling (detector) piece (Theorem 7.1, DERIVED-CONDITIONAL), while the locator is the adelic piece — unsolved externally and outside the corpus's present completion (which has only finite truncated-Euler-product prime structure, a detector). The realization framing is retired; ZS-M4's detector claim stands. F24 engages the correct frontier object, with anti-numerology tripwires pre-registered. No claim is made on RH, GRH-for-K, or determinant convergence.

Acknowledgements and Code Availability

Developed within the Z-Spin Cosmology corpus with AI assistance for symbolic and numerical exploration, the Nielsen–Olesen BVP, and drafting; the author is responsible for all claims. The vortex BVP (scipy.solve_bvp), the profile/tail diagnostics, the fluctuation-potential comparison, the scale-cylinder identities, and the z^* / α_{BK} anchors (Appendix A) were computed at machine and 30-digit precision and are reproducible. The frontier survey cites peer-reviewed work; crank and preprint-mill claims were filtered out by reliability criteria, not counted.

Appendix A. Verification Listing

A.1 NO vortex ($n = 1$), dimensionless $f'' + f'/x - (1-a)^2 f/x^2 - (\beta/2)(f^2-1)f = 0$; $a'' - a'/x + (1-a)f^2 = 0$; $f(0)=a(0)=0$, $f(\infty)=a(\infty)=1$. solve_bvp converged for $\beta = 1, 2$. f range $[0,1]$, monotone; core exponent $p = 0.996$; tail $1-f \sim K_0(x)$ (slope -1.06 , relRMS 0.019; power-fit RMS 0.33).
A.2 affine rescue: $f \approx \frac{1}{2}(1+\tanh(\Lambda \ln(x/R)))$ best fit $\Lambda = 1.015$, $R = 0.872$, RMS 0.022, max residual 0.077; tails differ (exp vs power).
A.3 fluctuation potential $U(x) = (1-a)^2/x^2 + (\beta/2)(3f^2-1)$: core $\approx +10^6$ ($\sim 1/x^2$), $x = 12 \rightarrow +\beta$. Prolate normal form $V_{Jac} = -\frac{1}{4} \tan^2 \xi - \frac{1}{2} \rightarrow -\infty$ at $\xi = \pm\pi/2$. $(3f^2-1)$ vs $(2f-1)^2$: best fit RMS 0.42.
A.4 scale cylinder: $\tau = \Lambda \ln(r/R) \Rightarrow r dr = (r^2/\Lambda) d\tau$; $W_Z = (2\pi\Lambda q)^2 (dq/d\tau) = 4\pi^2 \Lambda^5 \chi^2 (1-\chi^2)$; $(\Lambda/r^2) W_Z = V_{req}$ (SymPy diff = 0); $W_Z d\tau \propto \frac{1}{3} d(\chi^3)$.
A.5 anchors: $z^* = i^{\wedge}\{z^*\} = 0.43828294 + 0.36059247 i$; $|z^*| = 0.5675552$; $-\ln|z^*| = 0.5664173 = y^*\pi/2$; $|z^*|^2 = 0.3221189 = e^{\wedge}\{-y^*\pi\}$ (lock L3). Multiplier $|(i\pi/2)z^*| = 0.89151$ (Wilson margin, distinct).

Appendix B. Open Problems

O-F24.20. Determine whether any Z-core field (beyond the abelian-Higgs amplitude) can produce a $[-1,1]$ sign-changing profile with a power tail matching $\chi = \tanh$; absent that, Gate 1 stays closed.
O-F24.21 (frontier). The adelic/global completion (semilocal $\rightarrow$ global; “Weil cohomology”) — the locator step — unsolved externally; track CCM progress rather than re-deriving internally.
O-F24.22. Decide whether the corpus admits any adelic structure beyond the finite truncated Euler product; if not, the locator is permanently outside scope and F24 is a detector-only paper.

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Version History

v1.0–v1.4 (March 2026): A-washout; boundary-triple collapse; endpoint/spectrum separation; kinetic pullback & potential inertness; anchor-core BPS gate.

v2.0 (March 2026, terminus): Closes the arc. (i) Computed BPS-vortex no-go: the Nielsen–Olesen vortex (BVP-solved) does not realize the prolate well — profile $f \in [0,1]$ with exponential tail (not $\tanh$), and s-wave fluctuation potential with opposite endpoint sign and wrong bulk profile (Theorems 5.1–5.2). (ii) Scale-cylinder reframing resolves the angular tension but relocates rather than removes the coefficient barrier (Theorem 4.1). (iii) External frontier mapped: the detector→locator step is OPEN even for Connes–Consani–Moscovici with the same prolate operator; i-tetration = archimedean scaling/detector piece (Theorem 7.1, DERIVED-CONDITIONAL, via ZS-M4 Thm 3 $\alpha_{BK} = -\ln|z^*| = 0.566417 + \text{Connes decomposition}$); the locator is the adelic piece, OPEN externally and beyond the corpus's finite-Euler-product (detector) structure. Anti-numerology tripwires pre-registered. Realization framing retired; ZS-M4 detector claim retained. No RH proof claimed. Consolidated from internal Z-Spin Collaboration research notes.